

ISC Class XII Mathematics

Chapter 1 Relations and Functions

31 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Sections Covered

- I. Relations
- II. Functions – Domain and Range
- III. Types of Functions – One-One, Onto, Bijections
- IV. Composition of Functions
- V. Invertible Functions

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Type	Method / Pattern	★	Source(s)
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17	Counting functions / one-one / onto functions / bijections	Yes	Ex 1.3 Q10,12,13,14; Ex 1.3 Q29 (alt.)
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27	Proving one-one & onto, hence finding the inverse	Yes	Ex 1.5 Q19; Ex 1.5 Q20 (ISC 2019) (alt.)
28	Restricting the codomain to the range to make f invertible	Yes	Ex 1.5 Q24; Ex 1.5 Q25 (alt.)
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Formula Sheet

Section I – Relations: Counting

$$n(R) = 2^{n^2}$$

Number of relations on a set with n elements (n^2 possible ordered pairs, each either in or out of R)

Section I – Relations: Counting

$$n(R_{\text{reflexive}}) = 2^{n^2-n}$$

Number of reflexive relations on a set with n elements (n diagonal pairs are forced in; remaining $n^2 - n$ pairs are free)

Section I – Defining Properties

$$\text{Reflexive: } (a, a) \in R \forall a \in A$$

Every element must be related to itself

Section I – Defining Properties

$$\text{Symmetric: } (a, b) \in R \Rightarrow (b, a) \in R$$

Order of the pair can be reversed

Section I – Defining Properties

$$\text{Transitive: } (a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$$

A chain of two related pairs forces a third

Section I – Equivalence Relations

$$[a] = \{x \in A : (x, a) \in R\}$$

Equivalence class of a – the set of all elements related to a

Section II – Domain Restrictions

Even root: expression ≥ 0 ; Denominator $\neq 0$; $\log(\cdot)$:
argument > 0

The three most common domain-restricting conditions

Section II – Range (Auxiliary Angle Method)

$$a \sin x + b \cos x = r \sin(x + \alpha), \quad r = \sqrt{a^2 + b^2}$$

Put $a = r \cos \alpha$, $b = r \sin \alpha$; since $-1 \leq \sin(x + \alpha) \leq 1$, the range is $[-r, r]$

Section III – Counting Functions

Number of functions from $A \rightarrow B = n^m$

$m = |A|$, $n = |B|$ – each of the m elements of A has n independent choices

Section III – Counting Functions

$$\text{Number of one-one functions} = \begin{cases} {}^n P_m, & m \leq n \\ 0, & m > n \end{cases}$$

No repeated images allowed; impossible if A is bigger than B

Section III – Counting Functions

Onto functions $(n = 2) : 2^m - 2$; $(n = 3) : 3^m - 3(2^m - 1)$

Particular-case shortcuts for the general inclusion–exclusion onto-count formula

Section III – Counting Functions

$$\text{Number of bijections} = \begin{cases} m!, & m = n \\ 0, & m \neq n \end{cases}$$

Bijections only possible between equal-sized finite sets

Section IV – Composition

$$(g \circ f)(x) = g(f(x))$$

Apply f first, then g , to the result

Section IV – Composition

f, g one-one $\Rightarrow g \circ f$ one-one; f, g onto $\Rightarrow g \circ f$ onto

Injectivity and surjectivity are both preserved under composition

Section V – Invertibility

f is invertible $\iff f$ is bijective (one-one and onto)

The fundamental test before attempting to find f^{-1}

Section V – Finding the Inverse

Put $y = f(x)$, solve for x in terms of $y \Rightarrow f^{-1}(y) = x$
The standard algebraic method for finding f^{-1}

Section V – Invertibility

$$(f^{-1})^{-1} = f$$

Taking the inverse twice returns the original function

I. Relations

Type 1

Counting how many relations (or reflexive relations) are possible on a set of a given size. No solving needed – just correctly counting how many ordered pairs $A \times A$ has, and how many of those are "free choices."

Formula Used: $n(R) = 2^{n^2}$

Source: Exercise 1.1, Q1

Given: $A = \{-1, 1, 3\}$

Solution:

Step 1: $n(A) = 3$ – count the elements of A .

Step 2: $A \times A$ has $n^2 = 3^2 = 9$ ordered pairs – a relation on A is just any subset of $A \times A$.

Step 3: Each of the 9 pairs can independently be included or excluded from R , giving 2^9 possible subsets – this is the counting principle behind the formula.

Step 4: $2^9 = 512$.

Final Answer

Number of relations on $A = 512$

Common Mistake: Confusing $n(A \times A) = n^2$ with $n(A) = n$, and computing 2^n instead of 2^{n^2} .

Quick Recall: A relation is any subset of $A \times A$, so count is always 2 raised to (number of elements)²; for reflexive relations, the n diagonal pairs are forced in, so it's 2^{n^2-n} instead (see Q12: for $|A| = 3$, this gives 2^6).

Type 2

Finding the domain and/or range of a relation that is defined by a condition (set-builder form) rather than listed explicitly – you must find every valid input/output pair satisfying the condition.

Source: Exercise 1.1, Q2(i)

Given: $R = \{(x, y) : x + 2y = 8\}$ is a relation on $\mathbf{N}$

Solution:

Step 1: Rewrite the condition as $x = 8 - 2y$ – this lets us test each natural number value of y in turn.

Step 2: Try $y = 1, 2, 3, \dots$ and keep only those giving $x \in \mathbf{N}$: $y = 1 \Rightarrow x = 6$; $y = 2 \Rightarrow x = 4$; $y = 3 \Rightarrow x = 2$ – each is a valid pair since $x \in \mathbf{N}$.

Step 3: Try $y = 4$: $x = 8 - 8 = 0$, which is rejected since $0 \notin \mathbf{N}$ (here $\mathbf{N} = \{1, 2, 3, \dots\}$) – this is the boundary check that stops the list.

Step 4: Collect all valid y -values used – these form the range of R .

Final Answer

Range of $R = \{1, 2, 3\}$

Common Mistake: Forgetting to stop the list once y produces $x \leq 0$, or including 0 as a natural number when the book's convention is $\mathbf{N} = \{1, 2, 3, \dots\}$.

Quick Recall: Rearrange the condition to isolate one variable, then plug in valid values of the other one by one until you leave the required number system.

Type 3

Checking whether a given relation R on a set A is reflexive, symmetric, and/or transitive. This is tested by directly checking the three defining conditions against the pairs of R (or a hunting for a counter-example to disprove a property). This is the single most frequently tested relations skill.

★ **Board Favorite** – This exact check (R/S/T on an explicit or condition-defined relation) is asked in nearly every ISC paper as either a VSA or a full LA question – the book itself tags several of these questions directly to ISC Specimen Papers 2024 and 2025, and to past NCERT sources.

Formula Used: Reflexive: $(a, a) \in R \forall a \in A$

Formula Used: Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$

Formula Used: Transitive: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$

Source: Exercise 1.1, Q11

Given: $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (3, 2)\}$

Solution:

Step 1: **Reflexive check:** $A = \{1, 2, 3\}$ needs $(1, 1), (2, 2), (3, 3)$ all present – all three appear in R , so R is reflexive.

Step 2: **Symmetric check:** Take $(2, 1) \in R$ and ask if $(1, 2) \in R$ – it is not, so a single counter-example is enough to rule out symmetry.

Step 3: **Transitive check:** Take $(3, 2) \in R$ and $(2, 1) \in R$; transitivity would require $(3, 1) \in R$ – it is not present, so R fails transitivity too.

Final Answer

R is reflexive, but neither symmetric nor transitive

Common Mistake: Stopping after checking reflexivity and assuming symmetry/transitivity "look fine" without actually hunting for a counter-example pair.

Alternate Board-Style Example

Source: Exercise 1.1, Q14(i)

Given: $A = \{1, 2, \dots, 10\}$, $R = \{(x, y) : 2x - y = 0\}$

Solution:

Step 1: Since R is defined by a condition (not a roster list), first write out a few actual pairs: $x = 1 \Rightarrow y = 2$, $x = 2 \Rightarrow y = 4$, $x = 3 \Rightarrow y = 6$, $x = 4 \Rightarrow y = 8$, $x = 5 \Rightarrow y = 10$ – these are the only pairs in R , since $y = 2x \leq 10$ forces $x \leq 5$.

Step 2: **Reflexive check:** need $(1, 1) \in R$, i.e. $2(1) = 1$ – false, so R is not reflexive (one failure is enough).

Step 3: **Symmetric check:** $(1, 2) \in R$ but is $(2, 1) \in R$? That needs $2(2) = 1$, false – so R is not symmetric.

Step 4: **Transitive check:** $(1, 2) \in R$ and is there a $(2, z) \in R$? That needs $z = 4$, and $(2, 4) \in R$ is true; transitivity would then require $(1, 4) \in R$, i.e. $2(1) = 4$ – false, so R is not transitive.

Final Answer

R is neither reflexive, nor symmetric, nor transitive

Quick Recall: Test reflexive first (check every diagonal pair); for symmetric and transitive, actively hunt for ONE counter-example pair rather than trying to prove the property holds everywhere.

Type 4

Identifying or naming what kind of relation a small, explicitly-given relation is (or answering True/False on a stated property) – a quick recognition question rather than a full three-property table.

★ **Board Favorite** – Sourced directly from the ISC Specimen Question Paper 2024 – this "identify the relation" format is a recurring VSA/objective-type question.

Source: Exercise 1.1, Q5 (ISC Specimen Question Paper 2024)

Given: $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1)\}$

Solution:

Step 1: **Reflexive?** $(1, 1) \notin R$ – fails, so not reflexive.

Step 2: **Symmetric?** $(1, 2) \in R \Rightarrow (2, 1) \in R$ – true, and this is the only pair-and-reverse to check, so R is symmetric.

Step 3: **Transitive?** $(1, 2) \in R, (2, 1) \in R \Rightarrow$ transitivity needs $(1, 1) \in R$ – false, so not transitive.

Final Answer

R is neither reflexive nor transitive, but is symmetric

Common Mistake: Answering "identity relation" or "equivalence relation" from a vague first impression instead of actually running the three checks before naming the relation.

Quick Recall: "What kind of relation" questions are just the R/S/T check compressed into one line – run all three tests mentally before answering.

Type 5

Proving that a given relation is an equivalence relation (reflexive + symmetric + transitive together), and then finding the equivalence class of a specific element – the set of everything related to it.

★ **Board Favorite** – Equivalence-relation proofs are a standard long-answer favorite in every ISC Maths paper; the book explicitly tags this pattern as a Competency Focused Question, reflecting its weight in the current syllabus.

Formula Used: Equivalence relation: reflexive AND symmetric AND transitive

Formula Used: $[a] = \{x \in A : (x, a) \in R\}$

Source: Exercise 1.1, Q32 (Competency Focused Question)

Given: $A = \{5, 6, 7, 8, 9\}$, $R = \{(a, b) : |a - b| \text{ is divisible by } 2\}$

Solution:

Step 1: **Reflexive:** for any $a \in A$, $|a - a| = 0$, and 0 is divisible by 2 – so $(a, a) \in R$ for every a , giving reflexivity.

Step 2: **Symmetric:** if $|a - b|$ is divisible by 2, then $|b - a| = |a - b|$ is the same number, hence also divisible by 2 – so $(a, b) \in R \Rightarrow (b, a) \in R$.

Step 3: **Transitive:** suppose $|a - b|$ and $|b - c|$ are both even. Then $a - b$ and $b - c$ are both even integers (up to sign), so their sum $a - c = (a - b) + (b - c)$ is also even – hence $|a - c|$ is divisible by 2, giving $(a, c) \in R$.

Step 4: Since all three properties hold, R is an equivalence relation.

Step 5: **Equivalence class of 6:** find every $x \in A$ with $|x - 6|$ divisible by 2 – check each: $|5 - 6| = 1$ (odd, reject), $|6 - 6| = 0$ (even, keep), $|7 - 6| = 1$ (odd, reject), $|8 - 6| = 2$ (even, keep), $|9 - 6| = 3$ (odd, reject).

Final Answer

R is an equivalence relation; the elements related to 6 are $\{6, 8\}$

Common Mistake: Proving transitivity only for one specific numeric triple instead of arguing it for the general case (even + even = even), which is what a full proof requires.

Alternate Board-Style Example

Source: Exercise 1.1, Q37

Given: $A = \{1, 2, \dots, 10\}$, $(a, b) R (c, d) \iff a + d = b + c$ on $A \times A$

Solution:

Step 1: **Reflexive:** $(a, b) R (a, b)$ needs $a + b = b + a$, which is always true – so reflexive.

Step 2: **Symmetric:** if $a + d = b + c$, rearranging gives $c + b = d + a$, i.e. $(c, d) R (a, b)$ – so symmetric.

Step 3: **Transitive:** suppose $(a, b) R (c, d)$ i.e. $a + d = b + c$, and $(c, d) R (e, f)$ i.e. $c + f = d + e$. Adding these: $a + d + c + f = b + c + d + e$, which simplifies to $a + f = b + e$ – exactly the condition for $(a, b) R (e, f)$, so transitive.

Step 4: Hence R is an equivalence relation on ordered pairs.

Step 5: **Equivalence class of $(3, 4)$:** need all (c, d) with $3 + d = 4 + c$, i.e. $d = c + 1$ – so every pair one more in the second coordinate than the first.

Final Answer

$$[(3, 4)] = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8), (8, 9), (9, 10)\}$$

Quick Recall: Prove reflexive/symmetric/transitive in the general case using letters (not a single number example), then read off the equivalence class by testing the defining condition against every element of A .

Type 6

Constructing your own example of a relation that satisfies one specific combination of properties (e.g. symmetric but not reflexive or transitive) – a reverse-engineering question rather than checking a given relation.

Source: Exercise 1.1, Q33(i)

Given: Give an example of a relation which is symmetric but neither reflexive nor transitive.

Solution:

Step 1: Think of a real-world or geometric relation that naturally pairs things "both ways" but never relates something to itself – "is perpendicular to," on the set of all lines in a plane, is a natural candidate.

Step 2: Check symmetric: if line $\ell_1 \perp \ell_2$, then automatically $\ell_2 \perp \ell_1$ – so symmetric holds.

Step 3: Check not reflexive: a line is never perpendicular to itself, so $(\ell, \ell) \notin R$ for any ℓ – reflexivity fails, as required.

Step 4: Check not transitive: if $\ell_1 \perp \ell_2$ and $\ell_2 \perp \ell_3$, then ℓ_1 is actually *parallel* to ℓ_3 , not perpendicular – so transitivity fails, as required.

Final Answer

"Is perpendicular to" on the set of all lines drawn in a plane

Common Mistake: Picking a relation that accidentally satisfies one of the properties you were told to exclude (e.g. picking something that happens to also be transitive), without actually re-checking all three properties against your own example.

Quick Recall: Work backwards from real-world relations you already know the shape of (perpendicular, greater-than, sibling-of) rather than trying to invent an abstract set of pairs from scratch.

II. Functions – Domain and Range

Type 7

Directly evaluating a function at a given input, or solving $f(x) = k$ backwards to find x – straightforward substitution/algebra once the rule for f is known.

Source: Exercise 1.2, Q1

Given: $f(x) = \frac{3x^2 + 2x - 1}{x + 1}$, $x \in \mathbf{R}, x \neq -1$. Find $f(-3) + 1$.

Solution:

Step 1: Substitute $x = -3$ into the rule: $f(-3) = \frac{3(-3)^2 + 2(-3) - 1}{-3 + 1}$ – direct substitution, since $x = -3 \neq -1$ is inside the domain.

Step 2: Simplify numerator and denominator: $3(9) - 6 - 1 = 20$ on top, -2 on the bottom – arithmetic.

Step 3: $f(-3) = \frac{20}{-2} = -10$ – division.

Step 4: Add 1 as asked: $f(-3) + 1 = -10 + 1 = -9$.

Final Answer

$$f(-3)+1 = -9$$

Common Mistake: Substituting into the numerator and denominator separately but forgetting the requested "+1" at the end – always re-read exactly what final quantity is being asked for.

Quick Recall: Just substitute the given value directly into the formula, respecting the domain restriction, then finish the arithmetic asked for.

Type 8

Checking whether a given rule (often piecewise, or in unusual set-builder form) actually satisfies the definition of a function – every element of the domain must map to exactly one image.

★ **Board Favorite** – Well-definedness checks are a recurring conceptual trap in ISC papers – students often assume a rule is a function without checking for repeated or missing images.

Formula Used: A relation f is a function $\iff$ every element of the domain has exactly one image

Source: Exercise 1.2, Q2(i)

Given: $f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 2x, & 3 \leq x \leq 5 \end{cases}$

Solution:

Step 1: Notice the two pieces overlap at exactly $x = 3$ – this is the point to check carefully, since both branches are "active" there.

Step 2: Evaluate using the first branch: $f(3) = 3^2 = 9$.

Step 3: Evaluate using the second branch: $f(3) = 2(3) = 6$.

Step 4: Since $x = 3$ produces two different images (9 and 6), the rule violates the "exactly one image" requirement.

Final Answer

f is NOT a function, since the element 3 of the domain has two images, 9 and 6

Common Mistake: Only checking the two formulas individually for validity, without testing the shared boundary point where both pieces overlap.

Alternate Board-Style Example

Source: Exercise 1.2, Q2(ii)

Given: $R = \{(a, b) : \sqrt{a} = b\}$ on the set of real numbers

Solution:

Step 1: For R to be a function, every real a in the domain must give exactly one real b – test whether this holds for all reals, including negatives.

Step 2: Take $a = -4$: $\sqrt{-4}$ is not a real number – so $a = -4$ has no image at all under this rule.

Step 3: Since some real numbers (all negatives) fail to produce any real image, the rule is not defined for the whole of $\mathbf{R}$.

Final Answer

$\mathbf{R}$ is NOT a function on $\mathbf{R}$, since $\sqrt{a}$ is undefined (not real) for $a < 0$

Quick Recall: To disprove "this is a function," find one input that gives two images (or none at all) – always test boundary points of piecewise rules and negative/edge inputs of root or log rules first.

Type 9

Finding the domain and range of an algebraic function involving square roots, rational expressions, or absolute values – domain comes from restrictions (root ≥ 0 , denominator $\neq 0$), range comes from analysing what values the output can actually take.

★ **Board Favorite** – Domain-and-range of surd/rational/modulus functions is one of the most frequently tested VSA patterns in every ISC Maths paper, and appears repeatedly across NCERT Exemplar-tagged questions in the book.

Formula Used: For $\sqrt{g(x)}$ to be real: $g(x) \geq 0$

Source: Exercise 1.2, Q15(i) (Exemplar)

Given: $f(x) = \sqrt{16 - x^2}$

Solution:

Step 1: For the square root to be a real number, we need $16 - x^2 \geq 0$ – the defining domain condition for an even root.

Step 2: Rewrite as $x^2 \leq 16$, i.e. $-4 \leq x \leq 4$ – solving the quadratic inequality by factoring $16 - x^2 = (4 - x)(4 + x)$ and testing signs.

Step 3: So the domain is $[-4, 4]$.

Step 4: For the range: as x ranges over $[-4, 4]$, $16 - x^2$ ranges from 0 (at $x = \pm 4$) up to its maximum 16 (at $x = 0$) – so $16 - x^2 \in [0, 16]$.

Step 5: Taking square roots (which is increasing on non-negative numbers), $f(x) = \sqrt{16 - x^2} \in [0, 4]$.

Final Answer

Domain = $[-4, 4]$, Range = $[0, 4]$

Common Mistake: Writing the domain as $(-4, 4)$ (open interval) instead of $[-4, 4]$ – the endpoints ARE included since $16 - x^2 = 0$ is allowed (square root of zero is defined).

Alternate Board-Style Example*Source: Exercise 1.2, Q14(ii)*

Given: $f(x) = \frac{x^2 - 9}{x - 3}$

Solution:

Step 1: The denominator cannot be zero: $x - 3 \neq 0 \Rightarrow x \neq 3$ – so the domain excludes 3 only.

Step 2: Domain = $\mathbf{R} - \{3\}$.

Step 3: For the range, factor the numerator: $\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$ for all $x \neq 3$ – valid cancellation since $x - 3 \neq 0$ throughout the domain.

Step 4: So $f(x) = x + 3$ can take every real value EXCEPT the value it would take at the excluded point $x = 3$, namely $3 + 3 = 6$ – this value is never actually reached since $x = 3$ is not in the domain.

Final Answer

Domain = $\mathbf{R} - \{3\}$, Range = $\mathbf{R} - \{6\}$

Quick Recall: For domain: hunt for even roots (need ≥ 0) and denominators (need $\neq 0$); for range, simplify the expression fully first and track which single output value gets excluded by any cancelled factor.

Type 10

Finding the domain of a function whose formula involves a logarithm or a trigonometric expression in the denominator/argument – requires combining the standard log condition (argument > 0) or a trig inequality with the usual "denominator $\neq 0$ " check.

★ **Board Favorite** – Domain questions combining log or trig conditions are a recurring VSA pattern, since they test multiple domain rules (log positivity, non-zero denominator, trig range) in a single short question.

Formula Used: $\log(g(x))$ is defined $\iff g(x) > 0$

Source: Exercise 1.2, Q20(i)

Given: $f(x) = \log(4x - 3)$

Solution:

Step 1: For the logarithm to be defined, its argument must be strictly positive: $4x - 3 > 0$ – the fundamental log-domain rule.

Step 2: Solve: $4x > 3 \Rightarrow x > \frac{3}{4}$.

Final Answer

$$\text{Domain} = \left(\frac{3}{4}, \infty\right)$$

Common Mistake: Using ≥ 0 instead of > 0 for the log argument – $\log(0)$ is undefined, not zero.

Alternate Board-Style Example

Source: Exercise 1.2, Q21(ii)

Given: $f(x) = \frac{1}{\sqrt{1 - \cos x}}$

Solution:

Step 1: Since this is a square root in a denominator, we need the expression under the root to be strictly positive (not just ≥ 0 , since it also can't be zero or the fraction blows up): $1 - \cos x > 0$.

Step 2: This means $\cos x < 1$ – true for every real x except where $\cos x = 1$.

Step 3: $\cos x = 1$ exactly when $x = 2n\pi$ for integer n – recalling the standard solution set for $\cos x = 1$.

Step 4: So these specific points must be excluded from the domain.

Final Answer

$$\text{Domain} = \mathbf{R} - \{2n\pi : n \in \mathbf{Z}\}$$

Quick Recall: Combine every restriction visible in the formula at once – log needs its argument strictly positive, roots in denominators need their radicand strictly positive, and ordinary denominators just need to be non-zero.

Type 11

Finding the range of a function of the form $a \sin x + b \cos x$ (or with only one trig term) using the auxiliary angle (R-method): rewriting the expression as a single sine (or cosine) wave whose amplitude directly gives the range.

★ **Board Favorite** – The auxiliary-angle range-finding method is a classic, frequently repeated ISC pattern – it also resurfaces later in Calculus (maxima/minima of trig expressions), making it especially high-yield.

Formula Used: $a \sin x + b \cos x = r \sin(x + \alpha)$, where $r = \sqrt{a^2 + b^2}$, $a = r \cos \alpha$, $b = r \sin \alpha$

Source: Exercise 1.3, Q8(ii)

Given: $f(x) = 3 \sin x + 4 \cos x$

Solution:

Step 1: Here $a = 3$, $b = 4$. Compute $r = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ – this amplitude will directly become the bound on the range.

Step 2: Write $3 = 5 \cos \alpha$ and $4 = 5 \sin \alpha$ for some angle α – this is always solvable since $\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = 1$.

Step 3: Then $f(x) = 5 \cos \alpha \sin x + 5 \sin \alpha \cos x = 5 \sin(x + \alpha)$ – using the sine addition formula in reverse.

Step 4: Since $-1 \leq \sin(x + \alpha) \leq 1$ for every real $x + \alpha$, multiplying through by 5 gives $-5 \leq f(x) \leq 5$.

Final Answer

Range = $[-5, 5]$

Common Mistake: Assuming the range is simply $[-(a + b), a + b]$ (i.e. $[-7, 7]$ here) by just adding the coefficients, instead of computing the true amplitude $r = \sqrt{a^2 + b^2}$.

Alternate Board-Style Example

Source: Exercise 1.2, Q22(ii)

Given: $f(x) = 2 + 5 \sin 3x$

Solution:

Step 1: Here only one trig term is present, so no auxiliary angle is even needed – $\sin 3x$ itself already ranges over $[-1, 1]$ for all real x , since $3x$ still sweeps through all real values.

Step 2: Multiply through by 5: $5 \sin 3x \in [-5, 5]$.

Step 3: Add the constant 2 to every part of the inequality: $2 + 5 \sin 3x \in [2 - 5, 2 + 5] = [-3, 7]$.

Final Answer

$$\text{Range} = [-3, 7]$$

Quick Recall: Two trig terms together ($a \sin + b \cos$) $\Rightarrow$ amplitude is $\sqrt{a^2 + b^2}$; a single trig term with a constant added $\Rightarrow$ just shift the standard $[-1, 1]$ range by the coefficient and constant.

Type 12

Proving a functional equation – showing that f satisfies some algebraic identity (e.g. $f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right)$) by substituting the actual formula for f into both sides and simplifying until they match.

Source: Exercise 1.2, Q19

Given: $f(x) = \log\left(\frac{1+x}{1-x}\right)$. Show that $f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right)$

Solution:

Step 1: Start from the left-hand side and substitute the formula for f at both x and y :

$$f(x) + f(y) = \log\left(\frac{1+x}{1-x}\right) + \log\left(\frac{1+y}{1-y}\right).$$

Step 2: Use $\log A + \log B = \log(AB)$ to combine into a single logarithm: $= \log\left(\frac{(1+x)(1+y)}{(1-x)(1-y)}\right)$
– this is the key log law that makes the whole proof work.

Step 3: Now evaluate the right-hand side $f\left(\frac{x+y}{1+xy}\right)$ by substituting $\frac{x+y}{1+xy}$ in place of x in the original formula for f : $f\left(\frac{x+y}{1+xy}\right) = \log\left(\frac{1 + \frac{x+y}{1+xy}}{1 - \frac{x+y}{1+xy}}\right)$.

Step 4: Simplify the compound fraction by multiplying numerator and denominator by $(1+xy)$: numerator becomes $1 + xy + x + y = (1+x)(1+y)$, denominator becomes $1 + xy - x - y = (1-x)(1-y)$ – both factor nicely using grouping.

Step 5: So $f\left(\frac{x+y}{1+xy}\right) = \log\left(\frac{(1+x)(1+y)}{(1-x)(1-y)}\right)$, which is exactly what Step 2 gave for the left-hand side.

Final Answer

$$f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right), \text{ as required}$$

Common Mistake: Trying to prove the identity by picking specific numeric values of x and y instead of a general algebraic proof – a functional equation must be shown to hold for ALL x, y , not just verified for one pair.

Quick Recall: Substitute the actual rule into both sides separately, combine logs using the product law, and simplify the compound fraction on the RHS by clearing denominators – the two sides should collapse to the identical expression.

III. Types of Functions – One-One, Onto, Bijections

Type 13

Proving a function is one-one (injective) using the standard algebraic method: assume $f(x_1) = f(x_2)$ and show this forces $x_1 = x_2$.

Formula Used: f is one-one $\iff f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

Source: Exercise 1.3, Q1

Given: $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(x) = 3x - 2$

Solution:

Step 1: Assume $f(x_1) = f(x_2)$ for some x_1, x_2 in the domain – this is the starting assumption for any one-one proof.

Step 2: This gives $3x_1 - 2 = 3x_2 - 2$.

Step 3: Add 2 to both sides: $3x_1 = 3x_2$ – isolating the terms in x .

Step 4: Divide both sides by 3: $x_1 = x_2$ – this is exactly what one-one requires.

Final Answer

Since $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$, f is one-one

Common Mistake: Trying to "prove" one-one by testing a couple of specific numbers instead of the general algebraic argument – a finite number of checks can never prove injectivity for an infinite domain.

Quick Recall: Assume the two outputs are equal, then algebraically solve back to show the two inputs must also be equal.

Type 14

Proving a function is onto (surjective), or proving it is NOT onto by finding one element of the codomain with no preimage in the domain.

Formula Used: $f : A \rightarrow B$ is onto $\iff$ every $y \in B$ has some $x \in A$ with $f(x) = y$

Source: Exercise 1.3, Q2

Given: $f : \mathbf{N} \rightarrow \mathbf{N}$ defined by $f(x) = 2x - 1$

Solution:

Step 1: To disprove "onto," it's enough to find a single element of the codomain $\mathbf{N}$ that is never produced by f – pick an easy target like $y = 2$.

Step 2: Solve $2x - 1 = 2$ for x : $x = \frac{3}{2}$ – this is the value of x that WOULD produce $y = 2$.

Step 3: But $\frac{3}{2} \notin \mathbf{N}$, so there is no natural number x with $f(x) = 2$.

Step 4: Since $2 \in \mathbf{N}$ (the codomain) has no preimage in $\mathbf{N}$ (the domain), f fails to be onto.

Final Answer

f is NOT onto, since (for example) $2 \in \mathbf{N}$ has no preimage in $\mathbf{N}$

Common Mistake: Only checking that f produces some natural numbers, without noticing that $f(x) = 2x - 1$ only ever produces ODD numbers, so every even number in the codomain is automatically missed.

Quick Recall: To disprove onto, solve $f(x) = y$ for x in terms of y and show the result fails to lie in the domain for at least one choice of y .

Type 15

Checking injectivity (one-one) and surjectivity (onto) together for standard functions like x^2 or x^3 , across different choices of domain and codomain ($\mathbf{N}$, $\mathbf{Z}$, $\mathbf{R}$, $\mathbf{Q}$) – the SAME formula can behave completely differently depending on which sets it maps between.

★ **Board Favorite** – This exact NCERT-sourced question (checking x^2 and x^3 across $\mathbf{N}$, $\mathbf{Z}$, $\mathbf{R}$, $\mathbf{Q}$) is one of the most repeated ISC/board patterns for this chapter, because it tests the same rule against multiple domain/codomain pairs in one go.

Formula Used: f injective $\iff f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; f surjective $\iff \text{range}(f) = \text{codomain}$

Source: Exercise 1.3, Q18 (NCERT)

Given: Check injectivity and surjectivity of: (i) $f : \mathbf{N} \rightarrow \mathbf{N}, f(x) = x^2$ (ii) $f : \mathbf{Z} \rightarrow \mathbf{Z}, f(x) = x^2$ (iii) $f : \mathbf{R} \rightarrow \mathbf{R}, f(x) = x^2$ (iv) $f : \mathbf{N} \rightarrow \mathbf{Q}, f(x) = x^3$ (v) $f : \mathbf{Z} \rightarrow \mathbf{Z}, f(x) = x^3$ (vi) $f : \mathbf{R} \rightarrow \mathbf{R}, f(x) = x^3$

Solution:

Case: (i) $f : \mathbf{N} \rightarrow \mathbf{N}, f(x) = x^2$

Since $\mathbf{N}$ has only positive values, $f(x_1) = f(x_2) \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = x_2$ (as both are positive, no sign ambiguity) – so injective. But $2 \in \mathbf{N}$ has no natural square root, so not every codomain element is hit – not surjective.

Case: (ii) $f : \mathbf{Z} \rightarrow \mathbf{Z}, f(x) = x^2$

Here $f(2) = f(-2) = 4$, so two different inputs give the same output – not injective. Also $2 \in \mathbf{Z}$ has no integer square root – not surjective either.

Case: (iii) $f : \mathbf{R} \rightarrow \mathbf{R}, f(x) = x^2$

Same counter-example $f(2) = f(-2) = 4$ still applies – not injective. Also negative reals like -1 are never produced by x^2 – not surjective.

Case: (iv) $f : \mathbf{N} \rightarrow \mathbf{Q}, f(x) = x^3$

On $\mathbf{N}$, cubing is strictly increasing, so $x_1^3 = x_2^3 \Rightarrow x_1 = x_2$ – injective. But most rationals (e.g. $\frac{1}{2}$) are never a perfect cube of a natural number – not surjective.

Case: (v) $f : \mathbf{Z} \rightarrow \mathbf{Z}, f(x) = x^3$

Cubing is strictly increasing over all of $\mathbf{Z}$ too (unlike squaring, cubes preserve sign), so it remains injective. But e.g. $2 \in \mathbf{Z}$ has no integer cube root – not surjective.

Case: (vi) $f : \mathbf{R} \rightarrow \mathbf{R}, f(x) = x^3$

Still injective (strictly increasing over all reals). And now every real number DOES have a real cube root – so it is also surjective this time.

Final Answer

(i) injective, not surjective (ii) neither (iii) neither (iv) injective, not surjective (v) injective, not surjective (vi) both injective and surjective (bijective)

Common Mistake: Assuming injectivity/surjectivity of x^2 or x^3 is a fixed property of the "formula" – it actually depends entirely on the stated domain and codomain, which must be re-checked every single time.

Alternate Board-Style Example

Source: Exercise 1.3, Q22

Given: State whether the following functions $f : \mathbf{R} \rightarrow \mathbf{R}$ are bijective: (i) $f(x) = 2x + 1$ (ii) $f(x) = 3 - 4x$ (iii) $f(x) = 1 + x^2$

Solution:

Case: (i) $f(x) = 2x + 1$

$f(x_1) = f(x_2) \Rightarrow 2x_1 + 1 = 2x_2 + 1 \Rightarrow x_1 = x_2$, so injective. For any $y \in \mathbf{R}$, solving $2x + 1 = y$ gives $x = \frac{y-1}{2} \in \mathbf{R}$, so surjective too – bijective.

Case: (ii) $f(x) = 3 - 4x$

Similarly, $f(x_1) = f(x_2) \Rightarrow -4x_1 = -4x_2 \Rightarrow x_1 = x_2$, so injective; solving $3 - 4x = y$ gives $x = \frac{3-y}{4} \in \mathbf{R}$ always, so surjective – bijective.

Case: (iii) $f(x) = 1 + x^2$

$f(1) = f(-1) = 2$, so not injective. Also negative values like $y = 0$ are never produced since $1 + x^2 \geq 1$ always – not surjective either. Not bijective.

Final Answer

(i) Bijective (ii) Bijective (iii) Not bijective

Quick Recall: Linear functions $(ax + b, a \neq 0)$ on $\mathbf{R} \rightarrow \mathbf{R}$ are always bijective; even-power functions like x^2 are never injective on a domain that includes both signs, and never surjective onto all of $\mathbf{R}$.

Type 16

Checking one-one/onto for a piecewise-defined function (a different rule for different cases, e.g. odd vs. even inputs) – requires checking each case separately, and also checking across the two cases for injectivity conflicts.

★ **Board Favorite** – Piecewise bijection questions test case-based reasoning on top of the standard one-one/onto checks, making them a recurring higher-order pattern in ISC papers.

Source: Exercise 1.3, Q27

Given: $f : \mathbf{I} \rightarrow \mathbf{I}$ defined by $f(x) = \begin{cases} 2x + 1, & x \text{ odd} \\ 2x - 1, & x \text{ even} \end{cases}$

Solution:

Case: Finding $f(3)$

3 is odd, so use the first branch: $f(3) = 2(3) + 1 = 7$.

Case: Finding $f(4)$

4 is even, so use the second branch: $f(4) = 2(4) - 1 = 7$.

Case: Finding x such that $f(x) = -9$

Try the odd branch: $2x + 1 = -9 \Rightarrow x = -5$, and -5 is indeed odd – consistent, so this is a valid solution. Try the even branch: $2x - 1 = -9 \Rightarrow x = -4$, and -4 is indeed even – also consistent, so this is a second valid solution.

Case: Is f one-one?

From the very first two sub-cases, $f(3) = f(4) = 7$ – two DIFFERENT inputs give the SAME output, which is exactly the definition of failing one-one.

Final Answer

$f(3) = 7, f(4) = 7, x = -5$ or -4 when $f(x) = -9$; f is NOT one-one

Common Mistake: Checking each branch (odd/even) for injectivity separately and concluding "one-one" without ever comparing an odd-branch output against an even-branch output for a clash.

Alternate Board-Style Example

Source: Exercise 1.3, Q28(i)

Given: $f : \mathbf{N} \rightarrow \mathbf{N}$ defined by $f(n) = \begin{cases} n+1, & n \text{ odd} \\ n-1, & n \text{ even} \end{cases}$

Solution:

Case: One-one

Assume $f(n_1) = f(n_2)$. If both n_1, n_2 are odd: $n_1 + 1 = n_2 + 1 \Rightarrow n_1 = n_2$. If both are even: $n_1 - 1 = n_2 - 1 \Rightarrow n_1 = n_2$. If one is odd and the other even, one output is odd+1=even and the other is even-1=odd – so an odd input and an even input can never produce equal outputs, ruling out any cross-case clash. Hence one-one in every case.

Case: Onto

Take any $m \in \mathbf{N}$. If m is odd, then $m = n - 1$ for the even number $n = m + 1$, giving $f(n) = n - 1 = m$. If m is even, then $m = n + 1$ for the odd number $n = m - 1$, giving $f(n) = n + 1 = m$ (note $m - 1 \geq 1$ still natural since m even and $m \geq 2$). So every natural number has a preimage.

Final Answer

f is both one-one and onto (bijective)

Quick Recall: For piecewise functions, check one-one within each case first, then explicitly check whether an output from one case can ever collide with an output from the other case (usually resolved by a parity/sign argument), before checking onto case by case too.

Type 17

Counting problems: the number of functions, one-one functions, onto functions, or bijections between two finite sets, using the standard combinatorial formulas rather than solving anything about a specific rule.

★ **Board Favorite** – These counting formulas are asked almost every year as quick VSA/objective questions – being able to instantly pick the right formula (n^m , nP_m , or the onto shortcut) is high-value for time management.

Formula Used: Number of functions from $A(m) \rightarrow B(n) = n^m$

Formula Used: Number of one-one functions = nP_m ($m \leq n$)

Formula Used: Number of onto functions ($n = 2$) : $2^m - 2$; ($n = 3$) : $3^m - 3(2^m - 1)$

Source: Exercise 1.3, Q10, Q12, Q13, Q14

Given: (Q10) $A = \{1, 2, 3\}, B = \{3, 5\}$; (Q12) $A = \{a, b\}, B = \{1, 3, 5\}$; (Q13) $A = \{a, b, c\}, B = \{p, q\}$; (Q14) $A = \{1, 2, 3, 4\}, B = \{-1, 3\}$

Solution:

Case: Q10: Number of functions from $A = \{1, 2, 3\}$ to $B = \{3, 5\}$

Here $m = |A| = 3$, $n = |B| = 2$. Each of the 3 elements of A independently has 2 choices of image, so the total is $n^m = 2^3 = 8$.

Case: Q12: Number of one-one functions from $A = \{a, b\}$ to $B = \{1, 3, 5\}$

Here $m = 2$, $n = 3$, and since $m \leq n$ a one-one function is possible: count is ${}^3P_2 = 3 \times 2 = 6$ – picking distinct images for a then b in order.

Case: Q13: Number of onto functions from $A = \{a, b, c\}$ to $B = \{p, q\}$

Here $m = 3$, $n = 2$, so use the $n = 2$ shortcut: $2^m - 2 = 2^3 - 2 = 8 - 2 = 6$ – total functions minus the 2 "constant" functions that miss one element of B entirely.

Case: Q14: Number of onto functions from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 3\}$

Here $m = 4$, $n = 2$, so $2^m - 2 = 2^4 - 2 = 16 - 2 = 14$.

Final Answer

Q10: 8 functions; Q12: 6 one-one functions; Q13: 6 onto functions; Q14: 14 onto functions

Common Mistake: Swapping m and n in the formulas (e.g. computing m^n instead of n^m , or using mP_n instead of nP_m) – always fix which set is the domain (size m) and which is the codomain (size n) before substituting.

Alternate Board-Style Example

Source: Exercise 1.3, Q29

Given: Find the number of all one-one functions from the set $\{1, 2, 3, 4\}$ to itself.

Solution:

Step 1: Here $m = n = 4$ (domain and codomain are the same finite set) – when $m = n$, a one-one function is automatically also onto, i.e. it is a bijection (a permutation of the set).

Step 2: The number of such bijections is $n! = 4! = 4 \times 3 \times 2 \times 1$.

Final Answer

24 one-one functions

Quick Recall: Total functions $= n^m$; one-one needs $m \leq n$ and equals nP_m ; onto uses the $n = 2$ or $n = 3$ shortcut formulas; and whenever $m = n$, one-one automatically means bijective, giving simply $m!$.

Type 18

A combined question that determines bijectivity of a given function, finds an appropriate codomain that makes it onto, or finds all preimages of a specific value – several sub-questions built around the same function.

Source: Exercise 1.3, Q33

Given: $f(x) = \sqrt{25 - x^2}$

Solution:

Case: (a) If f is onto, find the codomain

The domain must be $[-5, 5]$ for the root to be real (since $25 - x^2 \geq 0 \iff -5 \leq x \leq 5$). As x ranges over $[-5, 5]$, $25 - x^2$ ranges over $[0, 25]$, so $f(x) = \sqrt{25 - x^2}$ ranges over $[0, 5]$ – for f to be onto, the codomain must be exactly this range.

Case: (b) If the domain is $[-5, 5]$, is $f(x)$ one-one?

Check $f(3) = \sqrt{25 - 9} = \sqrt{16} = 4$ and $f(-3) = \sqrt{25 - 9} = 4$ as well – two different inputs (3 and -3) give the same output, so f is NOT one-one on $[-5, 5]$.

Case: (c) Find all values of a for which $f(a) = 4$

Solve $\sqrt{25 - a^2} = 4 \Rightarrow 25 - a^2 = 16 \Rightarrow a^2 = 9 \Rightarrow a = \pm 3$ – squaring both sides and solving the resulting quadratic.

Final Answer

(a) Codomain $= [0, 5]$ (b) No, f is not one-one (c) $a = -3$ or 3

Common Mistake: Forgetting the $\pm$ when solving $a^2 = 9$, and reporting only $a = 3$ instead of both roots.

Quick Recall: For a semicircle-type function like $\sqrt{r^2 - x^2}$, the domain is always $[-r, r]$ and it is symmetric, so it can never be one-one on the full domain – and the range (hence onto-codomain) is always $[0, r]$.

IV. Composition of Functions

Type 19

Finding the composite function gof or fog when f and g are both given as algebraic formulas – substitute the entire rule of the inner function into the outer function.

★ **Board Favorite** – Algebraic composition is one of the most-repeated question types for this section – the book tags examples of this exact pattern to both ISC 2019 and a Competency Focused Question.

Formula Used: $(g \circ f)(x) = g(f(x))$

Source: Exercise 1.4, Q9 (ISC 2019)

Given: (i) $f(x) = 2x - 1, g(x) = x^2 + 3$ (ii) $f(x) = x + 1, g(x) = |x|$ (iii) $f(x) = x^3, g(x) = 2x^2 + 1$ (iv) $f(x) = \sin x, g(x) = 5x^2$

Solution:

Case: (i) $f(x) = 2x - 1, g(x) = x^2 + 3$

$(g \circ f)(x) = g(f(x)) = g(2x - 1) = (2x - 1)^2 + 3$ – substitute the whole expression $2x - 1$ wherever x appears in g . Expanding: $(2x - 1)^2 + 3 = 4x^2 - 4x + 1 + 3 = 4x^2 - 4x + 4$. Similarly $(f \circ g)(x) = f(x^2 + 3) = 2(x^2 + 3) - 1 = 2x^2 + 5$.

Case: (ii) $f(x) = x + 1, g(x) = |x|$

$(g \circ f)(x) = g(x + 1) = |x + 1|$. And $(f \circ g)(x) = f(|x|) = |x| + 1$.

Case: (iii) $f(x) = x^3, g(x) = 2x^2 + 1$

$(g \circ f)(x) = g(x^3) = 2(x^3)^2 + 1 = 2x^6 + 1$. And $(f \circ g)(x) = f(2x^2 + 1) = (2x^2 + 1)^3$.

Case: (iv) $f(x) = \sin x, g(x) = 5x^2$

$(g \circ f)(x) = g(\sin x) = 5\sin^2 x$. And $(f \circ g)(x) = f(5x^2) = \sin(5x^2)$.

Final Answer

(i) $(gof)(x) = 4x^2 - 4x + 4, (fog)(x) = 2x^2 + 5$ (ii) $(gof)(x) = |x + 1|, (fog)(x) = |x| + 1$ (iii) $(gof)(x) = 2x^6 + 1, (fog)(x) = (2x^2 + 1)^3$ (iv) $(gof)(x) = 5\sin^2 x, (fog)(x) = \sin(5x^2)$

Common Mistake: Mixing up the order – computing $f(g(x))$ when $(gof)(x) = g(f(x))$ was asked. Always apply the RIGHT-most function of the notation first.

Alternate Board-Style Example*Source: Exercise 1.4, Q11 (Competency Focused Question)*

Given: $f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$

Solution:**Case: (i) $(f \circ g)(x)$**

$f(g(x)) = f(2x - 3) = (2x - 3)^2 + 3(2x - 3) + 1$ – substitute $2x - 3$ for every x in f .
 Expand: $(4x^2 - 12x + 9) + (6x - 9) + 1 = 4x^2 - 6x + 1$.

Case: (ii) $(g \circ f)(x)$

$$g(f(x)) = g(x^2 + 3x + 1) = 2(x^2 + 3x + 1) - 3 = 2x^2 + 6x + 2 - 3 = 2x^2 + 6x - 1.$$

Case: (iii) $(f \circ f)(x)$

$f(f(x)) = f(x^2 + 3x + 1) = (x^2 + 3x + 1)^2 + 3(x^2 + 3x + 1) + 1$ – substitute the whole expression for f into itself; this is left in this expanded-but-not-multiplied-out form since fully expanding is not required for the composition itself.

Case: (iv) $(g \circ g)(x)$

$$g(g(x)) = g(2x - 3) = 2(2x - 3) - 3 = 4x - 6 - 3 = 4x - 9.$$

Final Answer

(i) $4x^2 - 6x + 1$ (ii) $2x^2 + 6x - 1$ (iii) $(x^2 + 3x + 1)^2 + 3(x^2 + 3x + 1) + 1$ (iv) $4x - 9$

Quick Recall: $(g \circ f)(x) = g(f(x))$ means: work out $f(x)$ first, then plug that ENTIRE result into g wherever its variable appears – never evaluate g and f separately and then combine.

Type 20

Finding the composite of two functions that are each given explicitly as a finite set of ordered pairs – a purely mechanical "look up the arrow" process rather than algebra.

Source: Exercise 1.4, Q6

Given: $A = \{3, 4, 5, 6\}$, $B = \{13, 14, 15, 16\}$, $C = \{23, 24, 25\}$. $f = \{(3, 13), (4, 14), (5, 15), (6, 16)\}$, $g = \{(13, 23), (14, 23), (15, 24), (16, 25)\}$

Solution:

Step 1: $(g \circ f)(3)$: first find $f(3) = 13$, then find $g(13) = 23$ – so $3 \mapsto 23$.

Step 2: $(g \circ f)(4)$: $f(4) = 14$, then $g(14) = 23$ – so $4 \mapsto 23$.

Step 3: $(g \circ f)(5)$: $f(5) = 15$, then $g(15) = 24$ – so $5 \mapsto 24$.

Step 4: $(g \circ f)(6)$: $f(6) = 16$, then $g(16) = 25$ – so $6 \mapsto 25$.

Final Answer

$$g \circ f = \{(3,23), (4,23), (5,24), (6,25)\}$$

Common Mistake: Trying to look up g first and f second (i.e. computing $f \circ g$ by mistake) – always trace f 's arrow first for $g \circ f$.

Quick Recall: For ordered-pair functions, composition is just "follow the arrow through f , then follow the arrow through g " for each element of A in turn.

Type 21

Verifying that composition is generally NOT commutative – showing $g \circ f \neq f \circ g$ for a specific pair of functions by computing both and finding they differ.

Source: Exercise 1.4, Q12

Given: $f(x) = x^2$, $g(x) = x + 1$, $f, g : \mathbf{R} \rightarrow \mathbf{R}$

Solution:

Step 1: Compute $(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1$.

Step 2: Compute $(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2$.

Step 3: Compare the two results: $x^2 + 1$ versus $(x + 1)^2 = x^2 + 2x + 1$ – these differ by the term $2x$, so they are not the same function.

Step 4: A single value confirms it too: at $x = 1$, $(g \circ f)(1) = 1^2 + 1 = 2$ but $(f \circ g)(1) = (1 + 1)^2 = 4$ – clearly unequal.

Final Answer

$$g \circ f \neq f \circ g, \text{ since } x^2 + 1 \neq (x + 1)^2 \text{ in general}$$

Common Mistake: Assuming composition behaves like multiplication (which IS commutative) – composition of functions is generally order-dependent and must be checked, not assumed.

Quick Recall: To disprove commutativity, compute both composites algebraically (or just test one convenient numeric value) and show they don't match.

Type 22

Showing that two functions are mutual inverses of each other by proving $gof = fog = I$ (the identity function) – both composites simply return the original input unchanged.

Formula Used: $I_{\mathbf{R}}(x) = x$ for all x

Source: Exercise 1.4, Q13

Given: $f(x) = x + 1$, $g(x) = x - 1$, $f, g : \mathbf{R} \rightarrow \mathbf{R}$

Solution:

Step 1: $(g \circ f)(x) = g(f(x)) = g(x + 1) = (x + 1) - 1 = x$ – the $+1$ and -1 cancel out exactly.

Step 2: $(f \circ g)(x) = f(g(x)) = f(x - 1) = (x - 1) + 1 = x$ – again a direct cancellation.

Step 3: Both composites simplify to exactly x , which is the definition of the identity function $I_{\mathbf{R}}$.

Final Answer

$gof = fog = I_{\mathbf{R}}$, since both composites equal x

Common Mistake: Stopping after checking only ONE of gof or fog equals the identity – both directions must be verified to conclude the functions are true (two-sided) inverses of each other.

Quick Recall: If applying one function then the other (in either order) always returns exactly the original input, the two functions are mutual inverses.

Type 23

Proving the general theorem that composition preserves injectivity and surjectivity: if f and g are both one-one, then $g \circ f$ is one-one; if f and g are both onto, then $g \circ f$ is onto. A "prove that" style proof using the definitions directly, not a specific numeric function.

★ **Board Favorite** – Recurring proof-based long-answer question in ISC papers – boards frequently ask students to prove general properties of composition rather than just compute one.

Source: Text Example 20 (NCERT)

Given: $f : A \rightarrow B$ and $g : B \rightarrow C$ are both one-one. Show that $g \circ f : A \rightarrow C$ is also one-one.

Solution:

Step 1: Let $x_1, x_2 \in A$ be such that $(g \circ f)(x_1) = (g \circ f)(x_2)$ – the standard starting assumption for any one-one proof.

Step 2: By definition of composition, this means $g(f(x_1)) = g(f(x_2))$.

Step 3: Since g is given to be one-one, this forces $f(x_1) = f(x_2)$ – applying the one-one property of g first, treating $f(x_1)$ and $f(x_2)$ as the two inputs to g .

Step 4: Since f is also given to be one-one, this in turn forces $x_1 = x_2$.

Step 5: So $(g \circ f)(x_1) = (g \circ f)(x_2) \Rightarrow x_1 = x_2$, which is exactly the definition of $g \circ f$ being one-one.

Final Answer

$g \circ f$ is one-one, whenever both f and g are one-one

Common Mistake: Trying to prove this with a specific example function instead of the general definitions – a "show that" theorem question needs a proof valid for ANY one-one f, g , not a single illustration.

Alternate Board-Style Example

Source: Text Example 21 (NCERT)

Given: $f : A \rightarrow B$ and $g : B \rightarrow C$ are both onto. Show that $g \circ f : A \rightarrow C$ is also onto.

Solution:

Step 1: Take any arbitrary element $z \in C$ – to prove onto, we must show SOME element of A maps to this arbitrary z under $g \circ f$.

Step 2: Since $g : B \rightarrow C$ is onto, there exists some $y \in B$ with $g(y) = z$ – using the onto-ness of g .

Step 3: Since $f : A \rightarrow B$ is onto, there exists some $x \in A$ with $f(x) = y$ – using the onto-ness of f , applied to the y just found.

Step 4: Then $(g \circ f)(x) = g(f(x)) = g(y) = z$ – chaining the two preimages together.

Step 5: So for every $z \in C$, there exists $x \in A$ with $(g \circ f)(x) = z$, which is exactly the definition of onto.

Final Answer

$g \circ f$ is onto, whenever both f and g are onto

Quick Recall: These are "definition-chasing" proofs: start from the assumption on the composite, unpack it one function at a time using the given property of g first (outer function), then f (inner function).

V. Invertible Functions

Type 24

Finding the inverse of a function that is given explicitly as a finite set of ordered pairs – simply reverse every ordered pair.

Formula Used: $f^{-1} = \{(b, a) : (a, b) \in f\}$

Source: Exercise 1.5, Q1

Given: $f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$

Solution:

Step 1: Since f is one-one (all four second coordinates 5, 3, 0, -2 are distinct), f^{-1} exists – a finite function only has an inverse if every image is unique.

Step 2: Reverse each ordered pair: $(1, 5) \rightarrow (5, 1)$, $(2, 3) \rightarrow (3, 2)$, $(3, 0) \rightarrow (0, 3)$, $(4, -2) \rightarrow (-2, 4)$.

Final Answer

$$f^{-1} = \{(5, 1), (3, 2), (0, 3), (-2, 4)\}$$

Common Mistake: Forgetting to first check that f is one-one (no repeated second coordinates) before assuming an inverse exists.

Quick Recall: For a finite function given as ordered pairs, just flip every pair left-to-right to get the inverse.

Type 25

Finding the inverse of a function given by an algebraic (linear or rational) formula: set $y = f(x)$ and solve algebraically for x in terms of y .

★ **Board Favorite** – This is the core, most frequently tested inverse-finding method – the book tags examples directly to NCERT and ISC 2023, confirming its recurring board weight.

Formula Used: Set $y = f(x)$, solve for $x \Rightarrow f^{-1}(y) = x$

Source: Exercise 1.5, Q8 (NCERT)

Given: $f : \mathbf{R} - \left\{ -\frac{4}{3} \right\} \rightarrow \mathbf{R}, \quad f(x) = \frac{4x}{3x+4}$

Solution:

Step 1: Let $y = \frac{4x}{3x+4}$ – setting the output equal to y so we can solve for x .

Step 2: Cross-multiply: $y(3x+4) = 4x \Rightarrow 3xy + 4y = 4x$ – clearing the denominator.

Step 3: Collect all x -terms on one side: $3xy - 4x = -4y \Rightarrow x(3y - 4) = -4y$ – factoring out x .

Step 4: Divide both sides by $(3y - 4)$: $x = \frac{-4y}{3y - 4} = \frac{4y}{4 - 3y}$ – rearranging the sign for a cleaner form.

Final Answer

$$f^{-1}(y) = \frac{4y}{4 - 3y}$$

Common Mistake: Making a sign error while collecting the x -terms on one side (a very common slip when the x -coefficient ends up negative before the final division).

Alternate Board-Style Example

Source: Exercise 1.5, Q9 (ISC 2023)

Given: $f(x) = [4 - (x - 7)^3]^{1/5}$ is a real invertible function.

Solution:

Step 1: Let $y = [4 - (x - 7)^3]^{1/5}$ – set the output equal to y .

Step 2: Raise both sides to the power 5 to undo the fifth root: $y^5 = 4 - (x - 7)^3$.

Step 3: Rearrange to isolate the cube: $(x - 7)^3 = 4 - y^5$.

Step 4: Take the cube root of both sides: $x - 7 = (4 - y^5)^{1/3}$ – undoing the cube (a real cube root always exists, unlike an even root).

Step 5: Add 7 to both sides: $x = 7 + (4 - y^5)^{1/3}$.

Final Answer

$$f^{-1}(x) = 7 + (4 - x^5)^{1/3}$$

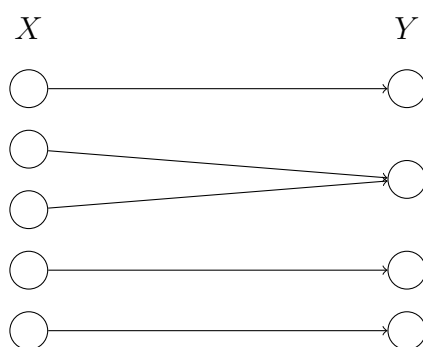
Quick Recall: Always work from the outermost operation inward: undo the outermost root/power first, then peel back each operation in reverse order, exactly like solving an onion of nested operations layer by layer.

Type 26

Determining whether a function is invertible directly from a mapping diagram or a graph, without any algebra – using the definition (one-one and onto) visually: multiple arrows into one point, or a vertical/horizontal line cutting the graph more than once, signal a failure.

★ **Board Favorite** – Reading invertibility off a diagram or graph (rather than computing it) is explicitly tested in recent ISC Specimen Papers (2024) as a quick conceptual VSA question.

Source: Exercise 1.5, Q10 (ISC Specimen Question Paper 2024)



Two elements of X map to the same element of Y , so f is not one-one and hence not invertible.

Given: $f : X \rightarrow Y$ is not invertible (mapping diagram given). State the reason.

Solution:

Step 1: Invertibility requires f to be bijective, i.e. both one-one and onto – this is the definition to check visually against the diagram.

Step 2: Looking at the diagram, two different elements of X (in the source book figure) send their arrows into the SAME element of Y .

Step 3: Since two distinct inputs share one output, the one-one condition fails – and a function that isn't one-one can never be invertible, regardless of whether it's onto.

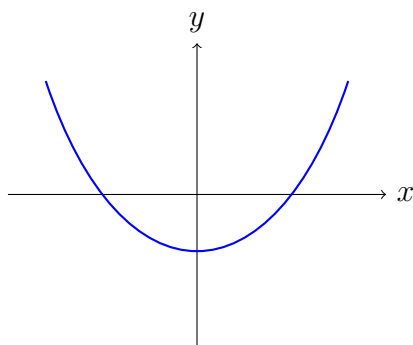
Final Answer

f is not invertible because f is not one-one

Common Mistake: Assuming a function fails to be invertible only because it's "not onto" without actually checking – always identify the SPECIFIC property (one-one or onto) that fails, since the reason must be stated precisely.

Alternate Board-Style Example

Source: Exercise 1.5, Q11 (ISC 2024)



Graph A: passes vertical line test

Graph A passes the vertical line test everywhere (every vertical line meets the curve at most once), so it represents a function of x . Graph B fails this test.

Given: Which one of the following graphs (Graph A or Graph B) is a function of x ?

Solution:

- Step 1: Apply the vertical line test: a curve represents a function of x only if every vertical line meets it at most once (each x can have at most one y).
- Step 2: In Graph A, the curve is a simple wave shape where every vertical line crosses it exactly once – so it passes the test.
- Step 3: In Graph B, the curve loops back on itself (an S-shape rotated so that some vertical lines cross it three times) – so a single x -value would correspond to multiple y -values, failing the test.

Final Answer

Graph A is a function of x

Quick Recall: One-one fails when two arrows/points share a target (or a horizontal line hits the graph twice); "is it even a function" fails when one input has two arrows/points leaving it (or a vertical line hits the graph twice) – read the diagram for exactly which failure is shown.

Type 27

The complete combined proof: showing an algebraic function is both one-one and onto (hence invertible), and then actually finding its inverse – the two skills from Type 15/25 chained together in one long-answer question.

★ **Board Favorite** – This combined "prove bijective, then find the inverse" question is the single most common long-answer format for this topic, with the book tagging examples directly to ISC 2019.

Source: Exercise 1.5, Q19

Given: $A = \mathbf{R} - \{-3\}$, $B = \mathbf{R} - \{2\}$, $f : A \rightarrow B$, $f(x) = \frac{2x+1}{x+3}$

Solution:

Step 1: **One-one:** assume $f(x_1) = f(x_2)$: $\frac{2x_1+1}{x_1+3} = \frac{2x_2+1}{x_2+3}$.

Step 2: Cross-multiply: $(2x_1+1)(x_2+3) = (2x_2+1)(x_1+3)$ – clearing both denominators at once.

Step 3: Expand both sides: $2x_1x_2 + 6x_1 + x_2 + 3 = 2x_1x_2 + 6x_2 + x_1 + 3$ – the $2x_1x_2$ and 3 terms are identical on both sides and cancel.

Step 4: This leaves $6x_1 + x_2 = 6x_2 + x_1 \Rightarrow 5x_1 = 5x_2 \Rightarrow x_1 = x_2$ – so f is one-one.

Step 5: **Onto:** let $y \in B$ be arbitrary and solve $y = \frac{2x+1}{x+3}$ for x : $y(x+3) = 2x+1 \Rightarrow xy + 3y = 2x+1 \Rightarrow x(y-2) = 1-3y \Rightarrow x = \frac{1-3y}{y-2}$.

Step 6: Since $y \neq 2$ (as $2 \notin B$), this value of x is always well-defined and real, and one can check $x \neq -3$ too (since $y \neq 2$ already excludes the case that would force $x = -3$) – so a valid preimage in A exists for every $y \in B$, confirming onto.

Step 7: Since f is both one-one and onto, f is invertible, and the formula found while proving onto IS the inverse (just relabel $y \rightarrow x$).

Final Answer

f is one-one and onto, hence invertible; $f^{-1}(x) = \frac{1-3x}{x-2}$

Common Mistake: Doing the onto proof correctly but forgetting that the SAME algebra already gives f^{-1} – there's no need for a separate calculation once "solve for x in terms of y " has been done for the onto proof.

Alternate Board-Style Example

Source: Exercise 1.5, Q20 (ISC 2019)

Given: $A = \mathbf{R} - \left\{\frac{8}{5}\right\}, \quad f : A \rightarrow A, \quad f(x) = \frac{8x+3}{5x-8}$

Solution:

Step 1: **One-one:** assume $f(x_1) = f(x_2)$, cross-multiply and expand exactly as before – the cross terms cancel, ultimately reducing to $x_1 = x_2$ (the algebra mirrors the main example's pattern).

Step 2: **Onto:** let $y \in A$ and solve $y = \frac{8x+3}{5x-8}$ for x : $y(5x-8) = 8x+3 \Rightarrow 5xy - 8y = 8x+3 \Rightarrow x(5y-8) = 8y+3 \Rightarrow x = \frac{8y+3}{5y-8}$.

Step 3: Notice this formula for x has exactly the same shape as f itself – this is the signature of a self-inverse function.

Step 4: Since a valid $x \in A$ exists for every $y \in A$ (as $y \neq \frac{8}{5}$ keeps the denominator non-zero), f is onto, hence bijective and invertible.

Final Answer

f is one-one and onto, hence invertible; $f^{-1}(x) = \frac{8x+3}{5x-8} = f(x)$ (f is self-inverse)

Quick Recall: Prove one-one by cross-multiplying $f(x_1) = f(x_2)$ and simplifying to $x_1 = x_2$; prove onto by solving $y = f(x)$ for x – and that very formula, with y relabelled as x , is automatically f^{-1} .

Type 28

When a function $f : \mathbf{N} \rightarrow \mathbf{N}$ (or similar) is one-one but NOT onto, it can be made invertible by restricting the codomain down to its actual range S – then $f : \mathbf{N} \rightarrow S$ becomes bijective and can be inverted.

★ **Board Favorite** – Restricting the codomain to the range (for a quadratic function on $\mathbf{N}$) is a recurring board pattern, often combined with a follow-up numeric evaluation of the inverse.

Formula Used: If $f : \mathbf{N} \rightarrow \mathbf{N}$ is one-one but not onto, $f : \mathbf{N} \rightarrow S$ (where $S = \text{range of } f$) is bijective

Source: Exercise 1.5, Q24

Given: $f : \mathbf{N} \rightarrow S, \quad f(x) = 4x^2 + 12x + 15, \quad S = \text{range of } f$

Solution:

Step 1: Rewrite the formula by completing the square: $4x^2 + 12x + 15 = (2x + 3)^2 + 6 -$ since $(2x + 3)^2 = 4x^2 + 12x + 9$, and $15 - 9 = 6$.

Step 2: **One-one:** assume $f(x_1) = f(x_2)$: $(2x_1 + 3)^2 = (2x_2 + 3)^2$. Since $x_1, x_2 \in \mathbf{N}$, both $2x_1 + 3$ and $2x_2 + 3$ are positive, so taking square roots (no sign ambiguity) gives $2x_1 + 3 = 2x_2 + 3 \Rightarrow x_1 = x_2$ – one-one.

Step 3: By definition, $f : \mathbf{N} \rightarrow S$ (codomain restricted to the actual range) is automatically onto – every value in S is, by definition of S , achieved by some input.

Step 4: So $f : \mathbf{N} \rightarrow S$ is bijective, hence invertible. To find the inverse, let $y = (2x + 3)^2 + 6$ and solve for x : $y - 6 = (2x + 3)^2 \Rightarrow \sqrt{y - 6} = 2x + 3$ (taking the positive root, since $2x + 3 > 0$ for $x \in \mathbf{N}$).

Step 5: So $x = \frac{\sqrt{y - 6} - 3}{2}$.

Final Answer

$$f^{-1}(y) = \frac{\sqrt{y - 6} - 3}{2}$$

Common Mistake: Taking both $\pm$ signs when square-rooting $(2x + 3)^2 = y - 6$ – since $x \in \mathbf{N}$ makes $2x + 3$ always positive, only the positive root is valid here.

Alternate Board-Style Example

Source: Exercise 1.5, Q25

Given: $f : \mathbf{N} \rightarrow S$, $f(x) = 9x^2 + 6x - 5$, $S =$ range of f . Find $f^{-1}(43)$ and $f^{-1}(163)$.

Solution:

Step 1: Complete the square: $9x^2 + 6x - 5 = (3x + 1)^2 - 6$ – since $(3x + 1)^2 = 9x^2 + 6x + 1$, and $-5 - 1 = -6$.

Step 2: Following the exact same method as the main example, $f : \mathbf{N} \rightarrow S$ is one-one (positive root only) and automatically onto its own range, hence invertible.

Step 3: Let $y = (3x + 1)^2 - 6$; solve for x : $y + 6 = (3x + 1)^2 \Rightarrow \sqrt{y + 6} = 3x + 1 \Rightarrow x = \frac{\sqrt{y + 6} - 1}{3}$ – so $f^{-1}(y) = \frac{\sqrt{y + 6} - 1}{3}$.

Step 4: Evaluate at $y = 43$: $x = \frac{\sqrt{49} - 1}{3} = \frac{7 - 1}{3} = 2$.

Step 5: Evaluate at $y = 163$: $x = \frac{\sqrt{169} - 1}{3} = \frac{13 - 1}{3} = 4$.

Final Answer

$$f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}; \quad f^{-1}(43) = 2, \quad f^{-1}(163) = 4$$

Quick Recall: Complete the square first to turn the quadratic into (linear)² + constant form – this instantly reveals both why it's one-one on $\mathbf{N}$ (positive root only) and how to invert it (undo the square, then solve the linear part).

Type 29

For a non-monotonic function (like a cubic with a local max and min), explaining why it fails to have an inverse over its whole domain, then restricting the domain to the LARGEST interval containing the point of symmetry where it does become invertible, and finding the resulting domain and range of f^{-1} .

★ **Board Favorite** – This restrict-the-domain-of-a-cubic pattern is explicitly tagged by the book as a Competency Focused Question – reflecting how the newer ISC syllabus (2026-27) leans into graph-reading, higher-order questions like this one. (*Only one worked example of this exact style is available in the current exercise set – let me know if you'd like a second one sourced from a specimen paper.*)

Source: Exercise 1.5, Q28 (Competency Focused Question)

Given: $f(x) = 2x^3 - 3x^2 - 12x + 8, \quad x \in \mathbf{R}$ (graph shown rising, then falling, then rising again)

Solution:

- Step 1: **(a) Why f has no inverse over all of $\mathbf{R}$:** the graph shown rises, turns and falls, then turns and rises again – so it is not strictly increasing (or decreasing) everywhere.
- Step 2: A horizontal line drawn through the "hump" region of the graph crosses the curve at three different points – meaning three different x -values give the same y -value there, so f is not one-one on all of $\mathbf{R}$, hence has no inverse.
- Step 3: **(b) Restricting the domain:** find the turning points by setting the derivative to zero: $f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x-2)(x+1)$, so $f'(x) = 0$ at $x = -1$ and $x = 2$ – these are exactly the local max/min points bounding the "hump".
- Step 4: For f to be invertible on the largest possible interval $[a, b]$ containing 0 (with $a < 0 < b$), we must stop exactly at these turning points – going even slightly beyond either one re-enters the non-monotonic hump. So $a = -1$ and $b = 2$.
- Step 5: Compute the endpoint values: $f(-1) = 2(-1) - 3(1) - 12(-1) + 8 = -2 - 3 + 12 + 8 = 15$; $f(2) = 2(8) - 3(4) - 12(2) + 8 = 16 - 12 - 24 + 8 = -12$.
- Step 6: Since f is decreasing on this restricted interval (moving from $x = -1$ to $x = 2$, the

function goes from 15 down to -12), the range of f (and hence the domain of f^{-1}) runs from the smaller value to the larger: $[-12, 15]$; the domain of f^{-1} equals the range of f restricted here, and the range of f^{-1} equals the restricted domain $[-1, 2]$.

Final Answer

Domain of $f^{-1} = [-12, 15]$, Range of $f^{-1} = [-1, 2]$

Common Mistake: Restricting the domain to some arbitrary "nice-looking" interval instead of using the actual turning points found from $f'(x) = 0$ – the interval must be bounded exactly at the points where monotonicity breaks down, and must be the LARGEST such interval containing the given point.

Quick Recall: Find where the derivative is zero – those are exactly the points bounding the largest interval on which a wavy graph becomes one-one; then swap domain and range to describe the inverse.

Type 30

Proving that a function is self-inverse, i.e. $f^{-1}(x) = f(x)$ for every x in the domain – applying f to itself should return the original input.

Formula Used: f is self-inverse $\iff f^{-1}(x) = f(x) \forall x$

Source: Exercise 1.5, Q29

Given: $f(x) = \frac{\pi^2}{x}$, $x \neq 0$, $x \in \mathbf{R}$

Solution:

Step 1: To show f is self-inverse, it's enough to show $f(f(x)) = x$ – if applying f twice returns the original input, f must be its own inverse.

Step 2: Compute $f(f(x)) = f\left(\frac{\pi^2}{x}\right) = \frac{\pi^2}{\pi^2/x}$ – substituting $\frac{\pi^2}{x}$ into f in place of x .

Step 3: Simplify the compound fraction: $\frac{\pi^2}{\pi^2/x} = \pi^2 \times \frac{x}{\pi^2} = x$ – dividing by a fraction means multiplying by its reciprocal.

Step 4: Since $f(f(x)) = x$ for every valid x , f is indeed self-inverse.

Final Answer

$f^{-1}(x) = f(x)$, so f is a self-inverse function

Common Mistake: Trying to verify this only at one specific numeric value of x instead of showing $f(f(x)) = x$ algebraically for a general x .

Quick Recall: To prove self-inverse, just compute $f(f(x))$ and check it simplifies back to plain x – functions of the form $\frac{k}{x}$ are always self-inverse this way.

Type 31

Verifying whether a proposed identity relating f and f^{-1} is actually true for a specific function – testing the claim algebraically rather than assuming it holds.

Source: Exercise 1.5, Q27 (Competency Focused Question)

Given: $f(x) = x^9$ on the board. Is $[f(x)]^{1/3} = [f^{-1}(x)]^3$ true?

Solution:

- Step 1: First find $f^{-1}(x)$: let $y = x^9$, so $x = y^{1/9}$, giving $f^{-1}(x) = x^{1/9}$ – taking the ninth root undoes the ninth power (valid here since odd roots are defined for all reals).
- Step 2: Compute the left-hand side of the claim: $[f(x)]^{1/3} = (x^9)^{1/3} = x^{9/3} = x^3$ – using the power-of-a-power rule.
- Step 3: Compute the right-hand side of the claim: $[f^{-1}(x)]^3 = (x^{1/9})^3 = x^{3/9} = x^{1/3}$ – again using the power-of-a-power rule.
- Step 4: Compare: LHS = x^3 but RHS = $x^{1/3}$ – these are NOT the same expression (e.g. at $x = 8$: LHS = 512, RHS = 2).

Final Answer

The claimed identity is FALSE: $[f(x)]^{1/3} = x^3 \neq x^{1/3} = [f^{-1}(x)]^3$; so Raghu's answer is incorrect

Common Mistake: Assuming any relationship "looks plausible" based on the general pattern of inverse-function notation, without actually computing both sides and comparing them.

Quick Recall: Never trust a proposed identity about f and f^{-1} by inspection – always compute f^{-1} properly first, then evaluate both sides of the claim independently and compare.